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| <b>POSITIONS</b>              | <b>Ross Assistant Professor</b><br>The Ohio State University                                                                                                                                                                                                                           | Fall 2025 - present        |
|                               | <b>Adjunct Professor</b><br>Cosumnes River College                                                                                                                                                                                                                                     | Spring 2018 - Spring 2019  |
| <b>EDUCATION</b>              | <b>Ph.D. in Mathematics</b><br>University of California, Santa Barbara                                                                                                                                                                                                                 | Fall 2019 - Spring 2025    |
|                               | <ul style="list-style-type: none"> <li>• Thesis: <i>Skein Theory of Affine ADE Subfactor Planar Algebras</i></li> <li>• Advisor: Stephen Bigelow</li> <li>• Certificate in College and University Teaching</li> </ul>                                                                  |                            |
|                               | <b>M.A. in Mathematics</b><br>University of Oklahoma                                                                                                                                                                                                                                   | Fall 2014 - Spring 2017    |
|                               | <b>B.S. in Mathematics</b><br>University of California, Davis                                                                                                                                                                                                                          | Fall 2010 - Spring 2014    |
| <b>PAPERS</b>                 | 5. <i>NIM-Representations of Tambara-Yamagami Generalizations</i> (with A. Czenky, E. McGovern, M. Müller, and A. Ros Camacho)<br>Submitted. arxiv: 2602.23502                                                                                                                         |                            |
|                               | 4. <i>A Mirror Deformation of Markov Numbers</i> (with L. Bittmann, P. Jouteur, E. Kantarcı Oğuz, and E. Yıldırım)<br>Submitted. arxiv: 2602.14802                                                                                                                                     |                            |
|                               | 3. <i>A Well-Defined Jellyfish Algorithm for the Affine <math>E_7</math> Subfactor Planar Algebra</i><br>Submitted. arxiv: 2601.09003                                                                                                                                                  |                            |
|                               | 2. <i>Skein Theory of Affine A Subfactor Planar Algebras</i><br>Submitted. arxiv: 2410.05519                                                                                                                                                                                           |                            |
|                               | 1. <i>A Determinant Formula of the Jones Polynomial for a Family of Braids</i> (with D. Asaner, S. Kumar, A. Pease, and A. Poudel), <i>Topology and its Applications</i> , <b>384</b> (2026).                                                                                          |                            |
| <b>RESEARCH INTERESTS</b>     | Quantum algebra, low-dimensional topology, operator algebras, tensor categories, diagrammatic algebras, subfactors, classical and virtual knot theory, quantum groups, higher categories, quantum topology, representation theory, $q$ -deformations of real numbers, cluster algebras |                            |
| <b>AWARDS</b>                 | USTARS Distinguished Graduate Student in Topology                                                                                                                                                                                                                                      | 2024                       |
|                               | Decolvenaere Research Accelerator Award, UCSB Graduate Division                                                                                                                                                                                                                        | 2023                       |
|                               | Individualized Professional Skills (IPS) Grant, UC Santa Barbara                                                                                                                                                                                                                       | 2023                       |
|                               | Math Department Graduate Fellowship, University of Oklahoma                                                                                                                                                                                                                            | 2015                       |
|                               | Richard V. Andree Memorial Scholarship, University of Oklahoma                                                                                                                                                                                                                         | 2015                       |
|                               | Yueh-Jing Lin Fund, UC Davis                                                                                                                                                                                                                                                           | 2013                       |
| <b>MENTORSHIP</b>             | The Ohio Summer Undergraduate Institute in Mathematics Mentor                                                                                                                                                                                                                          | June 2026 - August 2026    |
|                               | UCSB Directed Reading Program Mentor                                                                                                                                                                                                                                                   | December 2019 - June 2023  |
|                               | REU Graduate Mentor, UCSB                                                                                                                                                                                                                                                              | June 2022 - August 2022    |
|                               | Women in STEM Mentor, UCSB                                                                                                                                                                                                                                                             | September 2019 - June 2022 |
| <b>INVITED RESEARCH STAYS</b> | Institute of Mathematics of the Polish Academy of Sciences,<br>The Simons Semester on "Quantum Groups and Operator Algebras"                                                                                                                                                           | May 2027 - June 2027       |
| <b>TALKS</b><br>*=INVITED     | *37. Moab Topology Conference; Utah State University, Moab                                                                                                                                                                                                                             | May 2026                   |
|                               | <i>A Well-Defined Jellyfish Algorithm for the Affine <math>E_7</math> Subfactor Planar Algebra</i>                                                                                                                                                                                     |                            |
|                               | *36. UNCW Algebra and Number Theory Seminar; University of North Carolina<br>Wilmington<br><i>NIM-Representations of Tambara-Yamagami Extensions</i>                                                                                                                                   | February 2026              |

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| *35. Banff International Research Station Workshop; New Developments in Tensor Categories (Virtual)<br><i>A Well-Defined Jellyfish Algorithm for Affine E Subfactor Planar Algebras</i>                                                          | January 2026  |
| *34. Joint Mathematics Meetings; Recent Progress in Quantum Topology Special Session; Walter E. Washington Convention Center<br><i>A Well-Defined Jellyfish Algorithm for Affine E Planar Algebras</i>                                           | January 2026  |
| *33. Joint Mathematics Meetings; Recent Developments by Women in Mathematical Physics Special Session; Walter E. Washington Convention Center<br><i>Fusion Categories Arising from Affine A Planar Algebras</i>                                  | January 2026  |
| *32. Banff International Research Station Workshop; Operator Algebraic Quantum Groups<br><i>A Well-Defined Jellyfish Algorithm for Affine E Subfactor Planar Algebras</i>                                                                        | December 2025 |
| *31. West Coast Operator Algebra Seminar; Pomona College<br><i>A Well-Defined Jellyfish Algorithm for Affine E Subfactor Planar Algebras</i>                                                                                                     | December 2025 |
| *30. European Quantum Algebra Lectures; Cardiff University and University of Nottingham (Virtual)<br><i>Skein Theory of Affine ADE Subfactor Planar Algebras</i>                                                                                 | November 2025 |
| *29. East Coast Operator Algebras Seminar; University of Nebraska - Lincoln<br><i>Subfactor Planar Algebras at Index 4</i>                                                                                                                       | October 2025  |
| 28. Young Mathematicians in C* Algebras; University of Southern Denmark<br><i>Subfactor Planar Algebras of Type Affine E7</i>                                                                                                                    | July 2025     |
| *27. AWM Research Symposium; Non-commutative algebras, tensor categories, and diagrams Special Session; University of Wisconsin-Madison<br><i>Diagrammatics of Index 4 Subfactor Planar Algebras</i>                                             | May 2025      |
| *26. AMS Spring Western Sectional Meeting; Knot Theory and Quantum Topology Special Session; California Polytechnic State University<br><i>Diagrammatics of Index 4 Subfactor Planar Algebras</i>                                                | May 2025      |
| 25. Banff International Research Station; Women in Noncommutative Algebra and Representation Theory 4<br><i>Planar Algebras and their Corresponding Categories</i>                                                                               | March 2025    |
| 24. Quantum Groups, Tensor Categories and Quantum Field Theory Workshop; University of Oslo<br><i>Index 4 Subfactor Planar Algebras and Their Corresponding Categories</i>                                                                       | January 2025  |
| *23. Joint Mathematics Meetings; Knots, 3-manifolds, and Their Invariants Special Session; Seattle Convention Center<br><i>A determinant formula of the Jones Polynomial for a family of braids</i>                                              | January 2025  |
| 22. Joint Mathematics Meetings; Advances in Operator Algebras Special Session; Seattle Convention Center<br><i>Using Diagrams to Describe Quantum Symmetries and Subfactors</i>                                                                  | January 2025  |
| *21. AMS Western Sectional Meeting; Graphical Calculus in Representation Theory and Low-Dimensional Topology Special Session; University of California, Riverside<br><i>Index 4 Subfactor Planar Algebras and Their Corresponding Categories</i> | October 2024  |
| 20. Quantum Groups and Representation Theory; North Carolina State University<br><i>Index 4 Subfactor Planar Algebras and their Corresponding Categories</i>                                                                                     | October 2024  |
| *19. Young Mathematicians in C* Algebras; University of Glasgow<br><i>Using Diagrams to Describe Quantum Symmetries of Subfactors</i>                                                                                                            | August 2024   |
| *18. Oberwolfach Seminar; Classification of C*-Algebras and Dynamics Seminar; Oberwolfach Research Institute for Mathematics<br><i>Nuclear Dimension of Cuntz-Pimsner Algebras of C*-Correspondences</i>                                         | May 2024      |

17. Banff International Research Station Workshop; Skew Braces, Braids, and the Yang-Baxter Equation; Banff International Research Station  
*Skein Theory of Subfactor Planar Algebras* May 2024
- \*16. USTARS Distinguished Graduate Student in Topology Speaker; Underrepresented Students in Topology and Algebra Research Symposium; University of Iowa April 2024  
*Using Diagrams to Describe Quantum Symmetries of Subfactors*
15. Graduate Student Topology and Geometry Conference; Michigan State University April 2024  
*Skein Theory for Index 4 Subfactor Planar Algebras*
14. Southeastern Lie Theory Workshop; Quantum Structures in Lie Theory; University of Virginia March 2024  
*Skein Theory for Index 4 Subfactor Planar Algebras*
13. Women in Mathematics in Southern California Symposium; Pomona College February 2024  
*Using Diagrams to Describe Quantum Symmetries of Subfactors*
- \*12. Joint Mathematics Meetings; Knots, Skein Modules, and Categorification Special Session; Moscone Center January 2024  
*Skein Theory of Index 4 Subfactor Planar Algebras*
- \*11. Claremont Topology Seminar; Pomona College November 2023  
*Skein Theory for Affine ADE Subfactor Planar Algebras*
- \*10. Twinned Conference on  $C^*$ -Algebras and Tensor Categories; Fields Institute; and streamed live at International Centre for Mathematical Sciences November 2023  
*Skein Theory for Affine ADE Subfactor Planar Algebras*
9. Young Mathematicians in  $C^*$ -Algebras; KU Leuven August 2023  
*Skein Theory of Subfactor Planar Algebras*
8. Great Plains Operator Theory Symposium; The Ohio State University May 2023  
*Skein Theory for Index 4 Subfactor Planar Algebras*
7. AMS Spring Western Sectional; Algebraic Structures in Knot Theory Special Session; Fresno State University May 2023  
*Diagrammatic Presentations of Affine Type A Subfactor Planar Algebras*
6. Moab Topology Conference; Utah State University, Moab May 2023  
*Skein Theory for Affine ADE Subfactor Planar Algebras*
5. Redbud Topology Conference; University of Oklahoma March 2023  
*Skein Theory of Affine Type A Subfactor Planar Algebras*
4. Underrepresented Students in Topology and Algebra Symposium; University of Washington March 2023  
*Diagrammatic Presentations of Index 4 Planar Algebras*
- \*3. Quantum Groups Seminar; Université de Lille & University of Oslo (Virtual) March 2023  
*Skein Theory for Affine ADE Subfactor Planar Algebras*
2. Tech Topology Conference; Georgia Institute of Technology December 2022  
*Diagrammatic Presentations of Index 4 Subfactor Planar Algebras*
1. Topology Students Workshop; Georgia Institute of Technology July 2022  
*Diagrammatic Presentations of Affine Type A Subfactor Planar Algebras*

**TEACHING**  
**\*=ONLINE**  
**COURSE**

**Instructor of Record - The Ohio State University**  
\*Math 1116 - Excursions in Mathematics  
Math 1131 - Calculus for Business

Fall 2025 - present

**Lead TA - UC Santa Barbara**

Provided professional development and training to new math department TAs

Fall 2022 - Spring 2023

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|                  | <b>Teaching Assistant - UC Santa Barbara</b><br>MATH 111ABC - Abstract Algebra<br>MATH 108AB - Introduction to Linear Algebra<br>MATH 8 - Transition to Higher Mathematics<br>*MATH 8 - Transition to Higher Mathematics<br>*MATH 6AB - Vector Calculus<br>MATH 34B - Calculus for Social and Life Sciences                                                                                                                                                                                                                                                                                                                                                               | Fall 2019 - Spring 2025                                                                                                                                                                                                                                                                             |
|                  | <b>Associate Instructor - UC Santa Barbara</b><br>MATH 4A - Linear Algebra with Applications<br>*MATH 6A - Vector Calculus I<br>*MATH 4A - Linear Algebra with Applications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Summer 2020, 2021, 2024                                                                                                                                                                                                                                                                             |
|                  | <b>Adjunct Mathematics Professor - Cosumnes River College</b><br>MATH 30 - Pre-Algebra<br>STAT 300 - Introduction to Statistics and Probability<br>MATH 20 - Arithmetic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Spring 2018 - Spring 2019                                                                                                                                                                                                                                                                           |
|                  | <b>Teaching Assistant - University of Oklahoma</b><br>MATH 1743 - Business Calculus I<br>MATH 1914 - Calculus 1 (Fast Track)<br>MATH 2924 - Calculus 2 (Fast Track)<br>MATH 1823 - Calculus 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Fall 2015 - Spring 2017                                                                                                                                                                                                                                                                             |
|                  | <b>Instructor of Record - University of Oklahoma</b><br>MATH 1823 - Calculus 1<br>MATH 1503 - College Algebra                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Fall 2014 - Summer 2015 & 2016                                                                                                                                                                                                                                                                      |
| <b>SERVICE</b>   | Operator Algebra Mentor Network Mentee Board Member<br>AMS 2025 Fall Eastern Virtual Sectional Special Session Co-Organizer<br>Topology Seminar Co-Organizer<br>AWM Student Chapter at UCSB, Secretary<br>AWM EvenQuads 3 Data Collector<br>Directed Reading Program at UCSB Co-Organizer<br>Graduate Quantum Algebra and Topology Seminar Co-Organizer, UCSB<br>AWM Chapter at UCSB Founder and President<br>Quantum Algebra and Topology Seminar Co-Organizer, UCSB<br>AWM Essay Contest Judge<br>Math Teacher's Circle - Oklahoma City Founder and Instructional Leader                                                                                                | September 2025 - present<br>February 2025 - October 2025<br>October 2024 - June 2025<br>June 2023 - June 2025<br>June 2024<br>December 2020 - July 2023<br>July 2023 - January 2024<br>August 2022 - June 2023<br>September 2021 - June 2022<br>January 2020 - March 2020<br>April 2015 - June 2017 |
| <b>WORKSHOPS</b> | <b>Institute of Science and Technology Austria (ISTA)</b><br>Women in Topology V<br><b>International Centre for Mathematical Sciences (ICMS)</b><br>Women in Mathematical Physics 3<br><b>Oberwolfach Research Institute for Mathematics (MFO)</b><br>Classification of $C^*$ -Algebras and Dynamics Seminar<br><b>Banff International Research Station (BIRS)</b><br>Operator Algebraic Quantum Groups<br>Women in Noncommutative Algebra and Representation Theory 4<br>Skew Braces, Braids, and the Yang-Baxter Equation Workshop<br>Women in Operator Algebras 3<br><b>American Institute of Mathematics (AIM)</b><br>Quantum Invariants and Low-dimensional Topology | January 2027<br>April 2026<br>May 2024<br>November 2025<br>March 2025<br>May 2024<br>June 2023<br>August 2023                                                                                                                                                                                       |

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| <b>TRAVEL<br/>GRANTS</b> | National Science Foundation Travel Grant      | 2025, 2024, 2024, 2023 |
|                          | AWM Symposium Travel Grant                    | 2025                   |
|                          | GSA Conference Travel Grant, UC Santa Barbara | 2025                   |
|                          | UCSB Doctoral Student Travel Grant            | 2024                   |
|                          | AMS Sectional Travel Grant                    | 2024                   |
|                          | GSA Conference Travel Grant, UC Santa Barbara | 2024                   |
|                          | Operator Algebra Mentor Network Grant         | 2023                   |